CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

May 2012

CL302 Structural Analysis-II

Date: 01.05.2012, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Differentiate between Statically Determinate and Indeterminate Structures. **[04]**

Q - 2 (a) A fixed beam AB of span 6m carries two point loads of 20 kN each at distance 1.5m from each end. Draw shear force and bending moment diagrams for the beam and find out point of contra-flexure if any. **[08]**

(b) Draw influence line diagrams for a propped cantilever beam of length 10m for (1) Reactions at A (2) Reaction at B (3) Moment at A. Calculate ordinates at 1m interval. **[09]**

OR

Q - 2 (a) A fixed beam AB of span 4.5m carries a gradually varying load from zero at A to 30 kN at B. Determine the fixed end moments and support reactions and draws shear force and bending moment diagrams. **[08]**

(b) A two span continuous beam ABC has span AB=8m and span BC=4m. Draw influence lines for Va, Vb and Vc. Calculate ordinates at 2m interval. **[09]**

Q - 3 (a) Explain Muller-Breslau Principle. **[03]**

(b) A cable supported at its ends 40m apart carries loads of 20kN, 10kN and 12kN at distances 10m, 20m and 30m from the left end. If the point on the cable where the 10kN load is supported is 13m below the end supports. Determine (1) the reactions at the supports (2) the tensions in the different parts of the cable (3) the total length of the cable. **[11]**

OR

(b) A loaded cord ACDEFB is supported at the ends having span 50m. Support B is 6m higher than support A. The dip of the cord at D is 7.5m below the left support A. Loads at point C, D, E, F are 30kN, 40kN, 24kN and 18kN respectively, with horizontal distances 10m, 20m, 30m and 40m from the left support. Determine (1) the reactions at the supports (2) the tensions in the different parts of the cord (3) the total length of the cable. **[11]**

SECTION – II

Q - 4 An axial pull of 50 kN is suddenly applied to a steel bar 2m long and 1000 mm^2 in cross section. If modulus of elasticity of steel is 200 kN/mm^2, find (1) maximum instantaneous stress (2) maximum instantaneous extension (3) Strain Energy (4) Modulus of Resilience. [07]

Q - 5 (a) The internal diameter of the cylinder of a hydraulic jack is 100mm. It is required to operate up to a pressure of 10MPa. If the yield strength of the cylinder material is 40MPa, calculate the wall thickness. [05]

(b) A cantilever beam ACB with fixed end at A has a span of 6m. Point C is at a distance 4m from point A. The beam is loaded with udl of intensity 20kN/m between A and C. A point load of 60kN is acting at point B. Find out slope and deflection using castigliano's first theorem. Take $EI=10*10^{13}$ $N.mm^2$. For span AC, I=2I and for span CB, I=I. [09]

OR

(b) A propped cantilever beam ACB has fixed support at A and roller support at B. Span AC=3m and CB=1m. It carries a point load of 16kN at point C. Calculate reactions at supports using castigliano's second theorem and draw SF and BM diagrams. [09]

Q - 6 (a) Analyse a propped cantilever beam ACDB with fixed support at A and roller support at B. Span AC=CD=DB=2m. It carries two point loads of 20kN at point C and 40kN at D. Use method of consistent deformation. Choose Vb as a redundant. Draw SF and BM diagrams. [09]

(b) Derive Lame's equations for a thick cylinder subjected to internal and external pressure. [05]

OR

(b) A thick cylinder with internal and external diameter 100mm and 200mm respectively, is subjected to simultaneous internal pressure of 30MPa and external pressure of 15 MPa. Calculate the maximum tensile and compressive hoop stresses in the cylinder. [05]

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